Unsupervised Classification for Multilook Polarimetric SAR Images via Double Dirichlet Process Mixture Model

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***Abstract*— This article proposes a hierarchical double Dirichlet process mixture model (DDPMM) for multilook polarimetric synthetic aperture radar (PolSAR) data unsupervised classification. Specifically, within the framework of product model (PM), an observed PolSAR data point can be factorized as the multiplication representation of a positive-scalar texture variable and a complex-Wishart-distributed speckle component. Based on this assumption, the polarization DPMM and texture DPMM in the proposed model are hierarchically established to characterize the polarimetric matrix and texture variable, respectively, thus yielding the generation procedure of the observation data to be learned sufficiently. Meanwhile, instead of sharing the same texture vector in many existing PM-based methods, each data point in DDPMM is associated with its own texture vector, which can be characterized as the weighted summation of several densities via texture DPMM rather than following a single distribution, such that the texture information can be fully and flexibly captured. In particular, dual local spatial constraints based on the statistical representations of polarization and texture spaces are also explored on the two DPMMs, allowing the local correlation to be adequately and dynamically incorporated. Moreover, all closed-form updates are derived with the variational Bayesian inference algorithm and the cluster number of the proposed model can be determined automatically. Experimental results on four real PolSAR datasets demonstrate the superiority of the proposed DDPMM to some state-of-the-art methods.**

***Index Terms*— Dirichlet process mixture model, polarimetric synthetic aperture radar (PolSAR), remote sensing technology, unsupervised classification, variational Bayesian inference.**

I. Introduction

AS AN advanced remote sensing technology, fully polarimetric synthetic aperture radar (PolSAR), which is developed from SAR, has the advantages of long-range, strong penetrating ability, and can work in all-weather and multiclimate conditions. By capturing the scattering echoes, PolSAR can effectively represent the physical properties and the scattering mechanism of various land covers. Compared with conventional SAR technique, the PolSAR data points obtained by combining various polarization channels take the form of matrices, which include a more abundant representation of the target's characteristics. Hence, the PolSAR observation data are better applied to diverse tasks, such as vegetation monitoring [1], ship detection [2], [3], land cover classification [4], [5], disaster management [6], and so on, which could be of great significance for both civilian and military fields.

As a fundamental research, PolSAR image classification not only outputs the labels for land covers but also can be regarded as the input of object detection as well as other interpretation tasks for providing discriminative features. Thus, it has been intensively investigated over the past few decades, from the unsupervised methods to the supervised ones. Despite obtaining better accuracy, the supervised methods (e.g., the polarimetric coding network [7], complex neural network [8], and other deep learning networks [9], [10], [11], [12]) need to be fed with lots of manually annotated samples and involve a time-consuming and expensive training process that limits their application potential. In contrast, the unsupervised methods that obtain better applicability without labels have gained rapid development in recent years [13], [14], [15]. Hence, this article concerns the unsupervised classification of PolSAR image, which is generally classified into two categories, i.e., scattering property [16], [17], [18], [19], and statistical modeling [20], [21], [22], [23].

Specifically, these scattering property-based classification usually employs the incoherent or coherent polarimetric decomposition techniques [24], [25], [26] to acquire the physical scattering features of the PolSAR observations, according to which a range of parameters are determined (like the scattering entropy H, anisotropy A, angle α, and so on). Subsequently, the H/A/α–Wishart classifier [27] or other clustering algorithms are introduced to accomplish the classification [28], [29]. However, the above decomposition techniques still have some problems. For example, suppose only one or a few decomposition methods are chosen to characterize scattering properties. The few types of extracted features may not be matched with all kinds of scattering mechanisms involved in the scenarios. On the contrary, selecting too many decomposition methods may also generate numerous redundant features

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irrelevant to the classification. Thus, despite these decomposition techniques extracting the multitudinous features, only the limited kinds of physical scattering properties of land covers are depicted, making them unsuitable for complex scenes [14].

On the other hand, the statistical modeling-based classification supposes that each PolSAR observation data are sampled from a certain statistical distribution (such as the complex Gaussian [30] or complex Wishart distribution [31]). Following this assumption, the Bayesian posterior classifier or the likelihood estimator is introduced to fulfill the classification [32]. Nevertheless, with the development of PolSAR technology, the acquired PolSAR image has higher resolution and the complexity of imaging scenarios is also increasing, which lead to the number of scatterers varying from each image resolution cell, thereby violating the assumption of fully developed speckle pattern and generating some heterogeneous areas [33]. In that case, the complex Gaussian and complex Wishart distributions cannot be flexible enough to exactly characterize the PolSAR scattering or covariance matrix, which may achieve the suboptimal classification performance in the heterogenous areas. To this end, the product model (PM), defined by factorizing observation data as the multiplication representation of a positive real-scalar texture variable and a complex Wishart distributed speckle component, is designed to interpret the PolSAR data physically. Following this structural factorization, the texture variable can measure the fluctuation degree of the limited scatterers' number in various solution cells. Consequently, by choosing different texture variables (e.g., the Gamma, inverse Gamma, generalized Gamma [34], and fisher texture variables), some advanced statistical distributions (i.e., the K [35], Wishart generalized Gamma [36], $\mathcal{G}_p^0$ [37], and Kummer-U [38] distributions) can be derived from the PM framework by marginalizing the texture variable to further process the heterogeneity.

Furthermore, since these advanced distributions fail to flexibly characterize the multi-peak, asymmetric, and heavy-tailed data, particularly in the complex scenes with multiple terrain types, many efforts have been undertaken to develop them to their finite mixture model (FMM) variants (i.e., KMM [39], $\mathcal{G}_p^0$MM [40], Kummer-UMM [41], and WGΓMM [42]), thus allowing them to accurately approximating more complex data distribution profiles. In that case, all terrain types are modeled via their own densities (a.k.a. components), and the complex distribution is approximated via the weighted summation of several simple densities, which makes the above FMMs obtain better performance for PolSAR image classification.

However, some challenges still exist.

1) For these FMMs, the component number is a crucial parameter involved in the initialization and estimation processes. Specifically, if the component number is selected with a relatively large value, the model complexity will significantly increase, while manually setting a small value of it may learn an inadequate representation of data distribution in view of ignoring too much local information of the distribution curve.
2) Since the texture variables are described by a single distribution and marginalized in each component, the same texture vector is shared by the whole PolSAR data points, such that the extraction of texture information is inflexible and insufficient.
3) Moreover, due to the fact that the marginalization operation generates some complex non-exponential-family terms (e.g., the Bessel function of K distribution, exponential function of WGΓ distribution, and Gamma functions in $\mathcal{G}_p^0$ and Kummer-UMM distributions), which along with the texture parameters inherited from the marginalized variables increase the difficulty of parameters' estimation. Hence, most of these advanced FMMs can only approximately determine model parameters by utilizing the expectation–maximization (EM) algorithm, such that they are sensitive to the initializations, slow to converge, and may be prone to over-fitting.
4) Owing to the neglect of local correlation, these FMMs can be easily affected via the speckle noise, which conceals the real scattering information and reduces the credibility of resulting labels to some extent.

To solve the above problems, a novel hierarchical double Dirichlet process mixture model (DDPMM) with variational Baysian inference is proposed for unsupervised classification of PolSAR images. The main contributions of this article are summarized as follows.

1) Derived from the PM, a generative DDPMM is designed to explicitly build the generation procedure of the PolSAR observations. By constructing a polarization DPMM and a texture DPMM, the proposed model can incorporate more discriminative information from the polarization and texture perspectives. Meanwhile, a variational inference (VI) algorithm is developed to achieve an effective learning of DDPMM with the closed-form update solutions.
2) In DDPMM, all PolSAR data points are associated with their own texture variables, which are represented as the weighted summation of several densities rather than following single texture distribution. Therefore, this strategy allows the texture information to be captured sufficiently and offers the additional flexibility to further deal with the heterogeneity in PolSAR image.
3) For eliminating the influence of speckle noise, dual spatial constraints are designed according to the statistical representations of polarization and texture properties, respectively. By integrating the dual spatial constraints into the proposed model, more context information from the polarization and texture perspectives is dynamically considered, yielding the smoother latent label variable.

The rest of this article is structured as follows. Section II is devoted to elaborate the proposed DDPMM with dual constraints in detail, and some preliminaries of PolSAR data, PM, and VI algorithm are also briefly introduced. Comprehensive experiments and analysis are conducted in Section III. And conclusively, Section IV concludes this article.

II. Methodology

This section starts with the introduction of PolSAR data and PM theory, and then elaborates the construction and generation process of the hierarchical double Dirichlet process mixture model (DDPMM). The parameter estimation via the

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VI algorithm of proposed model is derived in detail. Moreover, to eliminate the noise effect, the dual spatial constraints are designed to further incorporate the local correlation, which will be fully demonstrated in the final subsection.

A. PolSAR Data and PM

Different from the single-channel SAR system, PolSAR system measures the backscattering signal of terrain types with various combinations of polarization modes, typically including vertical polarization (v) and horizontal polarization (h). For observed PolSAR images, an arbitrary pixel can be represented as

$$\mathbf{S} = \begin{bmatrix} S_{hh} & S_{hv} \\ S_{vh} & S_{vv} \end{bmatrix} \tag{1}$$

where $\mathbf{S}$ is a complex scattering matrix, and each element S_{UV} is the scattering amplitude that contains the complex-valued backscattering coefficient of the terrain target transmitted (received) in U (V) polarization [43]. Following the assumption of monostatic reciprocity, $\mathbf{S}$ can be reshaped as a scattering vector $\mathbf{g} = [S_{hh}, \sqrt{2}S_{hv}, S_{vv}]^T$ according to $S_{hv} = S_{vh}$, where $[\cdot]^T$ indicates the transpose operation. Furthermore, to mitigate the effect of speckle noise, the multilook complex covariance (MCC) matrix is exploited by averaging covariance matrices among a local area, i.e.,

$$\mathbf{C} = \frac{1}{L}\sum_{i=1}^{L} \mathbf{g}_i \mathbf{g}_i^H \tag{2}$$

where $[\cdot]^H$ indicates the Hermitian transpose and L denotes the number of looks.

For homogeneous areas, the statistical characteristic of MCC matrix is commonly described with the complex Wishart distribution in (3), which can be regarded as the basis of the statistical modeling-based methods

$$\mathcal{W}\left(\mathbf{C}|L, \mathbf{\Sigma}^{-1}\right) = \frac{L^{Ld}}{\Gamma_d(L)} \frac{|\mathbf{C}|^{L-d}}{|\mathbf{\Sigma}|^{-L}} \exp\{-L\mathrm{tr}(\mathbf{\Sigma C})\}. \tag{3}$$

Here, $\mathrm{tr}(\cdot)$ and $|\cdot|$ represent the trace and determinant of matrix, respectively. d is the element number of scattering vector, and $\mathbf{\Sigma}^{-1}$ denotes the expectation of observation $\mathbf{C}$ [44]. $\Gamma(x) = \int_0^{+\infty} w^{x-1}\exp(-w)dw$ is the Gamma function, and $\Gamma_d(L) = \pi^{d(d-1)/2}\prod_{i=0}^{i=d-1}\Gamma(L-i)$. Nevertheless, with the advancement of PolSAR technology, the image resolution and the complexity of imaging scenes are all increasing, which result in the effective scatters' number varying from one resolution cell to another and thereby cause the spatial fluctuation. Therefore, the polarization-independent scene texture in the heterogeneous area has a large scale of variation, and simultaneously the Wishart distribution is not flexible enough to exactly characterize MCC matrix of PolSAR data. To address this issue, the PM, defined by factorizing an observation data as the multiplicative representation of a texture variable and a complex Wishart-distributed speckle component, is developed to physically interpret the PolSAR data, which is given by

$$\mathbf{C} = \mathbf{t} \cdot \mathbf{A} \tag{4}$$

where the positive real-scalar texture variable $\mathbf{t}$ is introduced to represent the natural spatial variation of the radar cross section. By choosing different distributions for $\mathbf{t}$, many commonly-used statistical distributions (or the components in FMMs) can be derived from (4) with marginalization operation $p(\mathbf{C}) = \int p(\mathbf{C}|\mathbf{t}) \cdot p(\mathbf{t})d\mathbf{t}$ for effectively dealing with the heterogeneous PolSAR data. However, some intractable non-exponential-family terms may also be generated in view of the marginalization, which increase the difficulty of estimating parameters and may obtain some infinite values, therefore seriously limiting their potential applications.

B. DDPMM Construction and Its Generation Process

As aforementioned earlier, the existing FMMs have some disadvantages. Naturally, the DPMM, which treats the component number as a variable and generates the mixing coefficient by sampling processes, can leverage more prior information, thus allowing to construct infinite components for describing arbitrary distribution. For example, [20] designs the complex Gaussian-DPMM (GDPMM) to characterize the PolSAR data, which performs well in the homogeneous areas. Nevertheless, the stick-breaking representation in DPMM also makes the model form more complex, and thereby the scattering matrix $\mathbf{S}$ is selected as the input form in GDPMM instead of the MCC matrix $\mathbf{C}$, which contains more abundant statistical information. Meanwhile, GDPMM cannot deal with the heterogeneous areas in view of the texture variable being not considered. Aiming at these questions, [45] establishes the textured DPMM (tDPMM) to describe the texture information involved in MCC matrix, where the texture variable is directly multiplied with the speckle component to simplify the estimation process. However, the texture variable is just described by a single distribution, leading to its insufficient learning of texture information and making its spatial constraint only be imposed on the polarization perspective. To deal with these problems, the DDPMM is constructed by assigning each data point with a unique texture vector, which, particularly, is regarded as a weighted mixture of several distributions by DPMM, thus allowing to flexibly explore the abundant texture information in heterogeneous regions. Moreover, without the marginalization operation, each term about texture variable in DDPMM is guaranteed to be included in the exponential family, which reduces the difficulty of parameter estimation.

Concretely, in the proposed DDPMM, we assume that each PolSAR observation is factorized as the multiplicative representation of a texture variable t_n and a Wishart-distributed speckle component A_n. With this knowledge of PM as well as the given form of complex Wishart distribution in (3), the speckle A_n can be substituted with C_n/t_n to obtain the distribution about C_n. Hence, the statistical property of an observed polarization data point is explicitly modeled as

$$p\left(C_n|t_n, \theta^{(C)}\right) = \frac{L^{Ld}|C_n|^{L-d}}{\Gamma_d(L)|\Sigma|^{-L} \cdot t_n^{Ld}} \exp\left\{-\frac{L\mathrm{tr}(\Sigma C_n)}{t_n}\right\} \tag{5}$$

where $\theta^{(C)} = \{\Sigma, L\}$ is the parameter sets about C_n involved in complex Wishart distribution. With (5), a polarization DPMM is constructed for learning the generation process of observation C_n, which could effectively explore the polarization

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information and automatically select an appropriate model structure. Simultaneously, for characterizing the abundant texture information, we further design a texture DPMM for sampling the texture variable instead of employing a single texture distribution, which provides the additional flexibility to learn texture information and thereby assists the modeling of polarimetric data to some extent. Besides, the marginalization operation can be omitted for avoiding the non-exponential-family terms in view of the texture variable being explicitly described with texture DPMM. The hierarchical construction of proposed DDPMM is formulated as

$$\text{Texture DPMM:} \quad \begin{cases} G^{(t)} \sim \mathrm{DP}\left(G_0^{(t)}, \alpha^{(\phi)}\right) \\ \theta_i^{(t)} \middle| G^{(t)} \sim G^{(t)} \\ t_n \middle| \theta^{(t)} \sim p\left(t_n; \theta^{(t)}\right) \end{cases} \tag{6}$$

$$\text{Polarization DPMM:} \quad \begin{cases} G^{(C)} \sim \mathrm{DP}\left(G_0^{(C)}, \alpha^{(v)}\right) \\ \theta_i^{(C)} \middle| G^{(C)} \sim G^{(C)} \\ C_n | t_n, \quad \theta^{(C)} \sim p\left(C_n | t_n, \theta^{(C)}\right) \end{cases} \tag{7}$$

where $\theta^{(t)}$ is the parameter sets about t_n, and $\theta^{(t)}$ and $\theta^{(C)}$ are generated from the distribution $G^{(t)}$ and $G^{(C)}$, respectively. The prior of $G^{(\cdot)}$ obeys a nonparametric Dirichlet process (DP), i.e., $\mathrm{DP}(G_0^{(\cdot)}, \Delta)$ with base distribution $G_0^{(\cdot)}$ and concentration parameter Δ. In such case, $G^{(\cdot)}$ satisfies the following condition, i.e.,

$$\left(G^{(\cdot)}(Q_1), \ldots, G^{(\cdot)}(Q_r)\right) \sim \mathrm{DP}\left(\Delta G_0^{(\cdot)}(Q_1), \ldots, \Delta G_0^{(\cdot)}(Q_r)\right) \tag{8}$$

where $\{Q_1, \ldots, Q_r, \ldots\}$ is an arbitrary finite partition, and the distribution on arbitrary measure Q_r corresponds to $\Delta G_0^{(\cdot)}(Q_r)$. Because $G^{(\cdot)}$ is discrete, multiple $\theta_i^{(\cdot)}$ can take on the same value simultaneously, and the texture variable or observed data is seen as a mixture representation. Meanwhile, by directly assigning a data point C_n with a weighted t_n rather than sharing the same texture vector $\{t_i\}_{i=1}^M$ in many methods, the texture information can be learned more accurately. After the texture information is fully captured by the texture DPMM, it along with the polarization information incorporated via polarization DPMM jointly facilitates the generation of PolSAR observations.

When applying the proposed DDPMM for classification task, the generative processes are further represented as

$$t_n | y_n, \quad \theta^{(t)} \sim p\left(t_n | y_n, \theta^{(t)}\right) \tag{9}$$

$$C_n | z_n, \quad t_n, \theta^{(C)} \sim p\left(C_n | z_n, t_n, \theta^{(C)}\right) \tag{10}$$

by introducing the latent category-indicator variables $\mathbf{Y} = \{y_n\}_{n=1}^N$ and $\mathbf{Z} = \{z_n\}_{n=1}^N$, where $y_n = \{y_{n1}, \ldots, y_{ni}, \ldots\}$ and $z_n = \{z_{n1}, \ldots, z_{ni}, \ldots\}$. The non-zero element in y_n (or z_n) indicates which component the texture variable t_n (or observed data point C_n) belongs to. If the mixing proportions in two DPMMs are known, y_n and z_n can be sampled from the following multinomial distributions as:

$$y_n | \boldsymbol{\pi}(\boldsymbol{\phi}) \sim \mathrm{Mul}(y_n; \boldsymbol{\pi}(\boldsymbol{\phi})) \tag{11}$$

$$z_n | \boldsymbol{\omega}(\mathbf{v}) \sim \mathrm{Mul}(z_n; \boldsymbol{\omega}(\mathbf{v})). \tag{12}$$

More importantly, since the distribution $G^{(t)}$ (or $G^{(C)}$) from DP is discrete, the stick-breaking construction is introduced to explicitly calculate the mixing coefficient $\boldsymbol{\pi}$ (or ω), which is given by

$$\pi_i(\phi) = \phi_i \prod_{l=1}^{i-1}(1 - \phi_i), \quad \phi_i \sim \mathrm{Beta}\left(\phi_i; 1, \alpha^{(\phi)}\right) \tag{13}$$

$$\omega_i(\mathbf{v}) = v_i \prod_{j=1}^{i-1}\left(1 - v_j\right), \quad v_i \sim \mathrm{Beta}\left(v_i; 1, \alpha^{(v)}\right). \tag{14}$$

Here, (12) and (14) [or (11) and (13)] depict the sampling procedure of mixing coefficient by breaking a "stick" with unit length (i.e., total unit mixture proportion) to several short sticks. For example, the length of each short stick π_i indicates the contribution of ith component in texture DPMM, where its value is calculated by breaking the remaining stick with a portion. ϕ_i is calculated by a Beta distribution, which indicates the length of the ith breaking-off. Thus, by the stick-breaking construction, the mixing coefficient π_i is guaranteed to be positive, and $\alpha^{(\phi)}$ has the ability to control the breaking-off portion. This is because when $\alpha^{(\phi)}$ is a large value, the remaining "stick" will be longer, yielding to a large number of components, and when $\alpha^{(\phi)}$ is a very small value, there is a shorter remaining "stick" in the next breaking-off step. Significantly, with the correlation $G^{(t)} = \sum_{i=1}^{\infty} \pi_i \delta_{\theta_i^{t*}}$, when $\alpha^{(\phi)} = 0$, $G^{(t)}$ becomes a single value rather than a distribution, and $G^{(t)} = G_0^{(t)}$ if $\alpha^{(\phi)}$ is infinite.

The complete generation procedure of DDPMM has been established as above, and the graph model is presented in Fig. 1. In our model, for the latent variable-based texture DPMM consisted of (6), (9), (11), and (13), the mixing coefficient $\boldsymbol{\pi}(\boldsymbol{\phi})$ for each component is determined according to the concentration parameter $\alpha^{(\phi)}$. With this knowledge, the hidden variable $\mathbf{Y}$ is obtained for assisting calculation of $\boldsymbol{\pi}$ and indicating the category of t_n. And then, the nth texture variable t_n is sampled from $p(t_n; \theta^{(t)})$, such that the texture information is integrated. Consequently, the texture DPMM provides the additional flexibility to characterize more heterogeneous areas and a chance to explore the local similarity from the texture perspective. Similarly, according to the speckle-derived distribution, (7), (10), (12), and (14) construct the polarization DPMM for learning the polarization information, and finally the observed data C_n is generated by the given mixing coefficient $\boldsymbol{\omega}$, concentration parameter $\alpha^{(v)}$, and the learned texture variable t_n. After the calculations of two DPMMs are fulfilled, the texture classification information incorporated in texture DPMM along with the polarimetric classification information in polarization DPMM can be integrated into the category-indicator variable for classifying. Due to leveraging the stick-breaking representations in two DPMMs, the proposed model is allowed to be calculated via the VI algorithm as well as other optimization-based algorithms, which is expounded in Section II-C.

C. Variational Inference for the DDPMM

VI is a typical Bayesian estimation algorithm that can effectively solve the complex distribution and alleviate the

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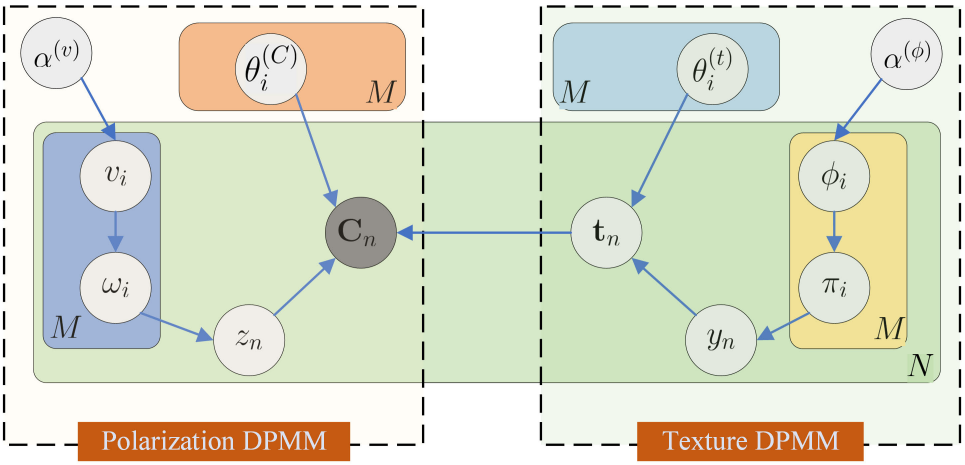


Fig. 1. Graph model of proposed DDPMM.

over-fitting phenomenon in the optimizing process, which has been widely used in DPMM, Gaussian processes, and other statistical frameworks. Specifically, considering that the posterior distribution in various statistical-modeling tasks generally has complicated forms or includes some intractable terms that is difficult to acquire directly, VI constructs a variational distribution involved in a relatively simple form for approximating this true posterior. And then, VI estimates the parameters of variational distribution rather than the original posterior. During the inference process, the Kullback–Leibler divergence (KLD) is a general choice to assess the approximation degree, i.e., $\mathrm{KLD}(q(\mathcal{Z})|p(\mathcal{Z}|\mathbf{C})) = \int q(\mathcal{Z})\ln(q(\mathcal{Z})/p(\mathcal{Z}|\mathbf{C}))d\mathcal{Z}$, where $p(\mathcal{Z}|\mathbf{C})$ denotes the true posterior, and $q(\mathcal{Z})$ denotes the constructed variational distribution. $\mathcal{Z} = \{\mathcal{Z}_j\}_{j=1}^{J}$ includes J variables required for estimation, and all variables are factorized as $q(\mathcal{Z}) = \prod_j q(\mathcal{Z}_j)$ by following the mean-field theory. Practically, by minimizing KLD to directly determine the optimal posterior is generally intractable. In that case, processing KLD is equivalent to optimizing the variational evidence lower bound (ELBO), which is seen as the objective function in VI, i.e.,

$$\max \mathcal{L}(q) = \int q(\mathcal{Z}) \ln \frac{p(\mathbf{C}, \mathcal{Z})}{q(\mathcal{Z})} d\mathcal{Z}. \tag{15}$$

Hence, the variational posterior is derived by (16), where $\mathbf{E}_{k\neq j}[\ln p(\mathbf{C}, \mathcal{Z})] = \int \ln p(\mathbf{C}, \mathcal{Z}) \prod_{k\neq j} q_k(\mathcal{Z}_k) d\mathcal{Z}_k$, and $\mathbf{E}[\cdot]$ denotes the expectation. It is worth noticing that VI can explicitly exploit the conjugacy between likelihood and prior, i.e., when assign the conjugate prior for each $\mathcal{Z}_j$, its variational posterior is guaranteed to keep the same form as prior, thus yielding a fast and efficient estimation

$$q^*(\mathcal{Z}_j) = \frac{\exp\{\mathbf{E}_{k\neq j}[\ln p(\mathbf{C}, \mathcal{Z})]\}}{\int \exp\{\mathbf{E}_{k\neq j}[\ln p(\mathbf{C}, \mathcal{Z})]\} d\mathcal{Z}_j}. \tag{16}$$

Following the VI strategy aforesaid, the variational learning algorithm is developed for the proposed DDPMM. Specifically, for keeping the conjugacy between prior and likelihood functions, some appropriate priors need to be assigned to the observation data and model parameters, i.e.,

$$p(C_n|z_{ni}=1, t_n, \Sigma_i) = \frac{L^{Ld}|C_n|^{L-d}}{\Gamma_d(L)|\Sigma_i|^{-L}\cdot t_n^{Ld}} \exp\left\{-\frac{L\mathrm{tr}(\Sigma_i C_n)}{t_n}\right\} \tag{17}$$

$$p(\Sigma_i; \eta_{i0}, W_{i0}^{-1}) = \frac{\eta_{i0}^{\eta_{i0}d}|\Sigma_i|^{\eta_{i0}-d}}{|W_{i0}|^{-\eta_{i0}}\Gamma_d(\eta_{i0})} \exp(-\eta_{i0}\mathrm{tr}(W_{i0}\Sigma_i)) \tag{18}$$

$$p(t_n|y_{ni}=1; a_{ni}, b_{ni}) = \frac{b_{ni}^{a_{ni}}}{\Gamma(a_{ni})} t_n^{-a_{ni}-1} \exp\left(-\frac{b_{ni}}{t_n}\right) \tag{19}$$

where L is seen as a constant, and the conditional density $p(C_n|z_{ni} = 1, t_n, \Sigma_i)$ characterizes ith component of observation data, which follows the same form in (10). Σ_i is sampled from the complex Wishart distribution with two hyperparameters η_i and W_i. More importantly, considering that the texture-dependent terms $\ln t$ and $1/t$ are included in the logarithmic form of (17), each t_n is assumed to follow an inverse Gamma distribution with hyperparameters a_{ni} and b_{ni}, thus avoiding the additional terms of t to be generated and yielding the conjugacy in VI being guaranteed.

Subsequently, to estimate the tractable solution of ELBO and the optimal posteriors for all random variables in DDPMM, the joint data log-likelihood function with the mean-field assumption in (15) is written as

$$\begin{aligned}\ln p(\mathbf{C}, \mathcal{Z}) = {} & \ln p(\mathbf{C}|\mathbf{Z}, \boldsymbol{t}, \boldsymbol{\Sigma}) + \ln p(t|\mathbf{Y}) + \ln p(\mathbf{Z}|\boldsymbol{\omega}(\mathbf{v})) \\ & + \ln p\left(\mathbf{v}; \alpha_0^{(v)}\right) + \ln p(\mathbf{Y}|\boldsymbol{\pi}(\boldsymbol{\phi})) \\ & + \ln p\left(\boldsymbol{\phi}; \alpha_0^{(\phi)}\right) + \ln p(\boldsymbol{\Sigma}; \theta_{i0})\end{aligned} \tag{20}$$

in which $\mathcal{Z} = \{\mathbf{Z}, \mathbf{t}, \mathbf{Y}, \boldsymbol{\Sigma}, \boldsymbol{\phi}, \mathbf{v}\}$ denotes the set of all variables that need to be estimated, and the true posterior is approximately represented by

$$p(\mathcal{Z}|\mathbf{C}) \approx q(\mathcal{Z}) = q(\mathbf{Y})q(\mathbf{t}|\mathbf{Y})q(\boldsymbol{\phi})q(\mathbf{Z})q(\mathbf{v})q(\boldsymbol{\Sigma}). \tag{21}$$

Here, since the texture variable is regarded as the weighted summation of several distributions in texture DPMM and is matched with the category indicator variable $\mathbf{Y}$, a conditional factorization between $\mathbf{t}$ and $\mathbf{Y}$ is reserved rather than considering a structural factorization between $\mathbf{t}$ and $\mathbf{Z}$.

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By incorporating these priors from (9)–(19) into (20), the log-likelihood is represented as follows, i.e.,

$$\begin{aligned}
&\ln p(\mathbf{C}, \mathcal{Z}) \\
&= \sum_n \sum_i z_{ni} \left\{ \ln \frac{L^{Ld}}{\Gamma_d(L)} + (L-d)\ln|C_n| - \frac{L}{t_n}\mathrm{tr}(\Sigma_i C_n) \right. \\
&\qquad \left. - Ld \ln t_n + L \ln|\Sigma_i| \right\} \\
&+ \sum_n \sum_i y_{ni} \left\{ \ln \frac{b_{ni0}^{a_{ni0}}}{\Gamma(a_{ni0})} - b_{ni0} \cdot \frac{1}{t_n} - (a_{ni0}+1)\ln t_n \right\} \\
&+ \sum_n \sum_i z_{ni} \ln \left\{ v_i \cdot \prod_{j=1}^{i-1} (1-v_j) \right\} \\
&+ \sum_n \sum_i y_{ni} \ln \left\{ \phi_i \cdot \prod_{l=1}^{i-1} (1-\phi_l) \right\} \\
&+ \sum_i \{(\eta_{i0} - d)\ln|\Sigma_i| - \eta_{i0}\mathrm{tr}(W_{i0}\Sigma_i)\} \\
&+ \sum_i \left(\alpha_0^{(v)} - 1\right) \ln(1-v_i) + \sum_i \left(\alpha_0^{(\phi)} - 1\right) \\
&\cdot \ln(1-\phi_i) + \mathcal{C}_O
\end{aligned} \tag{22}$$

where $\mathcal{C}_O$ represents the sum of all variable-irrelevant terms of $\mathcal{Z}$, and can be treated as a constant

$$\begin{aligned}
&\ln q^*(\mathbf{Z}) \\
&= \mathbf{E}_{\mathcal{Z}/\mathbf{Z}}\left[\ln p(\mathbf{C}, \mathcal{Z})\right] = \sum_n \sum_i z_{ni} \ln \rho_{ni}^{(Z)} + \mathcal{C}_O \\
&= \sum_n \sum_i z_{ni} \left\{ \ln \frac{L^{Ld}}{\Gamma_d(L)} + L\overline{\ln|\Sigma_i|} - Ld\overline{\ln t_n} + \overline{\ln v_i} \right. \\
&\qquad + \sum_{j=1}^{i-1} \overline{\ln(1-v_j)} - L\overline{t_n^{-1}}\mathrm{tr}\left(\overline{\Sigma_i} C_n\right) \\
&\qquad \left. + (L-d)\ln|C_n| \right\} + \mathcal{C}_O.
\end{aligned} \tag{23}$$

After substituting (22) into the (16) and following the variational strategy, the posterior distributions of all variables in (21) can be obtained, e.g., the posterior of $\mathbf{Z}$ with a logarithmic form can be given by (23), where $\overline{[\cdot]}$ refers to the expectation of $[\cdot]$ for writing and deriving expediently (i.e., $\overline{[\cdot]} = \mathbf{E}[\cdot]$). To keep the distribution form of the latent variable $\mathbf{Z}$ unchanging and normalize it, we obtain the following solution as:

$$\ln q^*(\mathbf{Z}) = \sum_{n=1}^{N} \sum_{i=1}^{M} z_{ni} \ln r_{ni}^{(Z)}. \tag{24}$$

Here, $r_{ni}^{(Z)} = \rho_{ni}^{(Z)} / \sum_{i=1}^{M} \rho_{ni}^{(Z)}$ is utilized to guarantee the summation of $r_{ni}^{(z)}$ over all i to be one and thereby ensuring the validity of $\mathbf{Z}$. The posterior distribution in (23) indicates the possibility of a specific data point C_n generated from the ith category in view of $\mathbf{E}[z_{ni}] = r_{ni}^{(Z)}$. Similarly, for the other indicator variable $\mathbf{Y}$ in texture DPMM, we have

$$\begin{aligned}
&\ln q^*(\mathbf{Y}) \\
&= \mathbf{E}_{\mathcal{Z}/\mathbf{Y}}\left[\ln p(\mathbf{C}, \mathcal{Z})\right] = \sum_n \sum_i y_{ni} \ln \rho_{ni}^{(Y)} + \mathcal{C}_O \\
&= \sum_n \sum_i y_{ni} \left\{ a_{ni0} \ln b_{ni0} - \ln\Gamma(a_{ni0}) - (a_{ni0}+1)\overline{\ln t_n} \right. \\
&\qquad \left. - b_{ni0}\overline{t_n^{-1}} + \overline{\ln \phi_i} + \sum_{l=1}^{i-1} \overline{\ln(1-\phi_l)} \right\} + \mathcal{C}_O
\end{aligned} \tag{25}$$

and its normalization form is given by (26), where $r_{ni}^{(Y)} = \rho_{ni}^{(Y)} / \sum_{i=1}^{M} \rho_{ni}^{(Y)}$

$$\ln q^*(\mathbf{Y}) = \sum_{n=1}^{N} \sum_{i=1}^{M} y_{ni} \ln r_{ni}^{(Y)}. \tag{26}$$

Consequently, the contribution of various texture components for generating t_n is determined by $\mathbf{E}[y_{ni}] = r_{ni}^{(Y)}$, and according to the cluster information over z_{ni} and y_{ni}, each observed data point can be labeled.

Analogously, by collecting all terms about a certain variable in ELBO and integrating the others, the variational posteriors of $\mathbf{v}$, $\boldsymbol{\phi}$, $\boldsymbol{\Sigma}$, and $\mathbf{t}$ can be respectively derived as (27)–(30)

$$\begin{aligned}
&\ln q^*(\mathbf{v}) \\
&= \sum_i \left\{ \left(\beta_i^{(v)} - 1\right) \ln v_i + \left(\alpha_i^{(v)} - 1\right) \ln(1-v_i) \right\} + \mathcal{C}_O \\
&= \sum_i \left\{ \left(\alpha_0^{(v)} + \sum_{j=1+i}^{\infty} \sum_n \overline{z_{nj}} - 1 \right) \ln(1-v_i) \right. \\
&\qquad \left. + \sum_n \overline{z_{ni}} \ln v_i \right\} + \mathcal{C}_O
\end{aligned} \tag{27}$$

$$\begin{aligned}
&\ln q^*(\boldsymbol{\phi}) \\
&= \sum_i \left\{ \left(\beta_i^{(\phi)} - 1\right) \ln \phi_i + \left(\alpha_i^{(\phi)} - 1\right) \ln(1-\phi_i) \right\} + \mathcal{C}_O \\
&= \sum_i \left\{ \left(\alpha_0^{(\phi)} + \sum_{l=1+i}^{\infty} \sum_n \overline{y_{nl}} - 1 \right) \ln(1-\phi_i) \right. \\
&\qquad \left. + \sum_n \overline{y_{ni}} \ln \phi_i \right\} + \mathcal{C}_O
\end{aligned} \tag{28}$$

$$\begin{aligned}
&\ln q^*(\boldsymbol{\Sigma}) \\
&= \sum_i \{(\eta_i - d)\ln|\Sigma_i| - \eta_i \mathrm{tr}(W_i \Sigma_i)\} + \mathcal{C}_O \\
&= \sum_i \left\{ \left(\eta_{i0} + L \sum_n \overline{z_{ni}} - d \right) \ln|\Sigma_i| \right. \\
&\qquad \left. - \mathrm{tr}\left[\left(\eta_{i0} W_{i0} + L \cdot \sum_n \overline{z_{ni}}\,\overline{t_n^{-1}} C_n \right) \Sigma_i \right] \right\} + \mathcal{C}_O
\end{aligned} \tag{29}$$

$$\begin{aligned}
&\ln q^*(\mathbf{t}|\mathbf{Y}) \\
&= \sum_n \sum_i -y_{ni} \left\{ (a_{ni}+1)\ln t_n + b_{ni} t_n^{-1} \right\} + \mathcal{C}_O
\end{aligned}$$

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$$= \sum_n \sum_i -\left[Ld\overline{z_{ni}} + \overline{y_{ni}}(a_{ni0}+1)\right]\ln t_n - \left[\overline{y_{ni}}b_{ni0} + \overline{z_{ni}}\cdot L\mathrm{tr}\left(\overline{\Sigma_i}C_n\right)\right]t_n^{-1} + \mathcal{C}_O. \quad (30)$$

Due to leveraging the conjugacy, the variational posteriors of these variables can maintain the same forms as their priors, i.e., the densities of $\boldsymbol{\phi}$ and $\boldsymbol{v}$ are still have the form of the Beta distributions, $\boldsymbol{\Sigma}$ is subject to the complex Wishart density, and the variable $\mathbf{t}$ follows the inverse Gamma density. Obviously, for (30), since the texture variable is not marginalized in the proposed DDPMM, the intractable terms about the texture variable (e.g., $|t|$, $t\ln t$, $\Gamma(t+\mathcal{C}_O)$, and so on) are not generated to influence its estimation. Hence, the reasonable assignment of texture prior makes the terms $\ln t$ and $1/t$ of ELBO be solved, which guarantees its posterior to be still an inverse Gamma distribution and the information included in the terms about $\ln t$ and $1/t$ in (19) to be incorporated in the inverse Gamma posterior for promoting the update of texture variable

$$\beta_i^{(v)} = \sum_n \overline{z_{ni}} + 1, \quad \alpha_i^{(v)} = \alpha_0^{(v)} + \sum_{j=1+1}^{\infty}\sum_n \overline{z_{nj}} \quad (31a)$$

$$\beta_i^{(\phi)} = \sum_n \overline{y_{ni}} + 1, \quad \alpha_i^{(\phi)} = \alpha_0^{(\phi)} + \sum_{l=i+1}^{\infty}\sum_n \overline{y_{nl}} \quad (31b)$$

$$\begin{aligned} a_{ni} &= a_{ni0} + Ld(\overline{z_{ni}}/\overline{y_{ni}}) \\ b_{ni} &= b_{ni0} + L(\overline{z_{ni}}/\overline{y_{ni}})\mathrm{tr}\left(\mathbf{C}_n\overline{\Sigma_i}\right) \\ \eta_i &= \eta_{i0} + L\sum_n \overline{z_{ni}} \end{aligned} \quad (31c)$$

$$W_i = 1\Big/\left\{\eta_{i0}W_{i0} + L\sum_n \overline{z_{ni}}\mathbf{C}_n\overline{t_n^{-1}}\right\}. \quad (31d)$$

With this knowledge, we could be straightforward to acquire each hyperparameter's update equation by (31a)–(31d) in an element-wise form, which provides the high convergence speed and avoids overfitting phenomenon. After that, the parameters of the proposed DDPMM are determined via alternatively evaluating (31a)–(31d) until reaching the stop criterion. Simultaneously, there are many essential expectations in (32) required to be calculated, according to which each $\mathcal{Z}_j$ can be updated during iterations by fixing other $\mathcal{Z}_{k,k\neq j}$, i.e.,

$$\begin{aligned} \mathbf{E}[z_{ni}] &= r_{ni}^{(Z)}, \quad \mathbf{E}\left[y_{ni}\right] = r_{ni}^{(Y)}, \quad \mathbf{E}[\Sigma_i] = W_i^{-1} \\ \mathbf{E}[\ln v_i] &= \psi\left(\beta_i^{(v)}\right) - \psi\left(\alpha_i^{(v)} + \beta_i^{(v)}\right), \quad \mathbf{E}\left[t_n^{-1}\right] = \frac{a_{ni}}{b_{ni}} \\ \mathbf{E}[\ln\phi_i] &= \psi\left(\beta_i^{(\phi)}\right) - \psi\left(\alpha_i^{(\phi)} + \beta_i^{(\phi)}\right) \\ \mathbf{E}[\ln t_n] &= \ln b_{ni} - \psi(a_{ni}) \\ \mathbf{E}[\ln(1-v_i)] &= \psi\left(\alpha_i^{(v)}\right) - \psi\left(\alpha_i^{(v)} + \beta_i^{(v)}\right) \\ \mathbf{E}[\ln|\Sigma_i|] &= \ln\left|W_i^{-1}\right| - d\ln\eta_i + \sum_{j=0}^{d-1}\psi(\eta_i - j) \\ \mathbf{E}[\ln(1-\phi_i)] &= \psi\left(\alpha_i^{(\phi)}\right) - \psi\left(\alpha_i^{(\phi)} + \beta_i^{(\phi)}\right) \end{aligned} \quad (32)$$

where ψ represents the digamma function with the form $\psi(x) = d\ln\Gamma(x)/dx$.

Once the proposed DDPMM fulfill the training process, the classification labels of the PolSAR data points are obtained via recognizing the largest element in z_n, which is given by

$$\mathrm{label}_n = \arg\max_i \mathbf{E}[z_{ni}]. \quad (33)$$

After reasonably initializing, the latent polarization and texture category information can be integrated by the two stick-breaking mixing proportions to some extent, which are transmitted to the latent variables, and subsequently, the latent variables could pass them to the label_n for classification.

Consequently, the proposed model improves the ability of statistical modeling by constructing an extra texture DPMM, where the texture mixture strategy allows our model to fit complex texture distribution and also provides a chance to impose the smoothness constraint of texture perspective. As such, the latent category-indicator variables include more discriminant classification information than the general DPMM.

D. Dual Constraints

Though the proposed DDPMM can address the heterogeneity, the inherent speckle noise in PolSAR image still critically affects the clustering procedure, resulting in learning the numerous misclassifications [47], [48], [49]. As such, some efforts in [32], [45], [50], [51], and [52] have been made to integrate the local spatial correlation as constraints into their algorithms for restraining a noise pixel by its neighboring pixels. With these constraints, the pairwise similarities between a pixel and its neighbors are calculated as the responsibility masks to reflect the prior knowledge over label spaces. However, these constraints are restricted to the discrete label or only imposed on the MCC matrix, which may incorporate the context information inadequately and not fully leverage the characteristics of PM.

To this end, the dual spatial constraints, which are specially designed based on the statistical representations of polarization and texture spaces, are embedded into the proposed model for dynamically involved in the parameter estimation, such that our model can fully capture the local correlation of the pairwise observation PolSAR data points and the pairwise texture variables.

Specifically, it is reasonable to consider that within a local block in a PolSAR image, pairwise pixels formed by the central pixel and its arbitrary neighboring pixel have the similar spatial correlation and belong to the same terrain type. Therefore, we assume that the polarimetric matrices and texture variables of the pairwise pixels also have the similar probabilistic representation, and two responsibility masks $\overrightarrow{h} = \{h_n\}_{n=1}^N$ and $\overrightarrow{p} = \{p_n\}_{n=1}^N$ are utilized to quantify the degree of the similarity relations, where each element in probability vector $h_n = [h_{n1}, \ldots, h_{ni}, \ldots]$ and $p_n = [p_{n1}, \ldots, p_{ni}, \ldots]$ represents the nth pixel and texture variable belonging to ith cluster, respectively, i.e.,

$$h_{ni} = \frac{\omega_i\, p(C_n)}{\sum_j \omega_j\, p(C_n)}, \quad p_{ni} = \frac{\pi_i\, p(t_n)}{\sum_l \pi_l\, p(t_n)} \quad (34)$$

where ω_i and π_i take the form of (13) and (14), respectively. Hence, the polarization and texture information of whole image are integrated into these two responsibility masks for classification. Based on them, the polarization and

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texture similarity measures of pairwise pixels n and m are given by

$$\mathcal{R}_{nm}^{(P)} = \sum_i z_{ni} \ln \frac{h_{ni}}{h_{mi}} + z_{mi} \ln \frac{h_{mi}}{h_{ni}}$$
$$\mathcal{R}_{nm}^{(T)} = \sum_i y_{ni} \ln \frac{p_{nk}}{p_{mi}} + y_{mi} \ln \frac{p_{mi}}{p_{ni}}. \tag{35}$$

Here, $\mathcal{R}_{nm}^{(P)}$ and $\mathcal{R}_{nm}^{(T)}$ are both the symmetric measures. When these similarity measures are designed as dual constraints, they can be integrated into the ELBO to obtain a new objective function with the form as (36), where $\mathcal{S}$ is a $Sz \times Sz$ local patch and includes N_S pixels. The first term in (36) incorporates the classification information to calculate each pixel's label, and the second term can be treated as a penalty term for encouraging all pixels in $\mathcal{S}$ to represent the same category

$$\widetilde{\mathcal{L}}(q) = \int q(\mathcal{Z}) \ln \frac{\widetilde{p}(\mathbf{C}, \mathcal{Z})}{q(\mathcal{Z})} d\mathcal{Z}$$
$$= \int q(\Lambda) \ln \frac{p(\mathbf{C}, \mathcal{Z})}{q(\mathcal{Z})} d\mathcal{Z}$$
$$- \sum_n \sum_{m \in \mathcal{S}} \frac{1}{2N_S} \cdot \int q(\Lambda) \left(\mathcal{R}_{nm}^{(P)} + \mathcal{R}_{nm}^{(T)}\right) d\mathcal{Z}. \tag{36}$$

By substituting $\ln \widetilde{p}(\mathbf{C}, \mathcal{Z})$ into (22), the dual constraints are embedded into the learning process of our DDPMM, which dynamically incorporate the local context, thus transforming the proposed model being a patch-level classification method. Following variational algorithm and structural factorization in (21), all variables can be estimated with the same strategy as (22)–(30). Hence, $\ln \rho_{ni}^{(Z)}$ and $\ln \rho_{ni}^{(Y)}$ of new ELBO with dual constraints are conveniently obtained as follows, i.e.,

$$\ln \rho_{ni}^{(Z)} = \ln \frac{L^{Ld}}{\Gamma_d(L)} + L\overline{\ln|\Sigma_i|} + \overline{\ln v_i} + \sum_{j=1}^{i-1} \overline{\ln(1 - v_j)}$$
$$- \frac{Ld}{N_S} \sum_{m \in \mathcal{S}} \overline{\ln t_m} - \frac{L}{N_S} \mathrm{tr}\left(\sum_{m \in \mathcal{S}} \overline{t_m^{-1}} C_m \overline{\Sigma_i}\right)$$
$$+ (L - d) \sum_{m \in \mathcal{S}} \frac{1}{N_S} \ln |C_m| \tag{37}$$

$$\ln \rho_{ni}^{(Y)} = a_{ni0} \ln b_{ni0} - \ln \Gamma(a_{nio}) + \overline{\ln \phi_i} + \sum_{l=1}^{i-1} \overline{\ln(1 - \phi_l)}$$
$$- (a_{ni0} + 1) \sum_{m \in \mathcal{S}} \frac{1}{N_S} \overline{\ln t_m} - b_{ni0} \sum_{m \in \mathcal{S}} \frac{1}{N_S} \overline{t_m^{-1}}. \tag{38}$$

Due to the fact that DPMM could determine the component number automatically according to the data, there may generate a problem that the determined number of components in polarization DPMM is not equal that in texture DPMM during a certain iteration. In such case, the two DPMMs have different cluster numbers, and it is difficult to directly associate one unique texture variable for a polarimetric component. Subsequently, these misaligned components will lead to the latent variables y_{ni} and z_{ni} must be summed as y_n and z_n before participating in other variables' updates, while y_{ni} and z_{ni} should directly participate the calculations under the same value of component numbers. Thus, the classification information cannot be correctly transmitted from the latent variables to each estimated variational parameter. To deal with this problem, a new $\ln \rho$ is generated to integrate polarization information and texture information during each iteration via $\ln \rho = \lambda \ln \rho_{ni}^{(Z)} + (1 - \lambda) \ln \rho_{ni}^{(Y)}$, and the $\ln \rho_{ni}^{(Z)}$ and $\ln \rho_{ni}^{(Y)}$ will be replaced by it for constraining two DPMMs to have the same number of components and aligning these components. Accordingly, with new $\ln \rho$, the classification information included in texture aspect is transferred from Y to Z in view of each polarimetric component being directly associated with a unique texture component, and the latent variable could directly participate the updating processes of other variables.

For the other hyperparameters, the update equations are redetermined as follows:

$$\beta_i^{(v)} = \sum_n \overline{z_{ni}} + 1, \quad \alpha_i^{(v)} = \alpha_0^{(v)} + \sum_{j=i+1}^{\infty} \sum_n \overline{z_{nj}} \tag{39a}$$

$$\beta_i^{(\phi)} = \sum_n \overline{y_{ni}} + 1, \quad \alpha_i^{(\phi)} = \alpha_0^{(\phi)} + \sum_{l=i+1}^{\infty} \sum_n \overline{y_{nl}} \tag{39b}$$

$$a_{ni} = a_{ni0} + \left[\frac{1}{N_S} + \sum_{m \in \mathcal{S}, m \neq n} \frac{1}{N_S} \overline{y_{mi}}\right] \frac{\lambda L d}{1 - \lambda}$$

$$b_{ni} = b_{ni0} + \mathrm{tr}\left(\left[\frac{1}{N_S} + \sum_{m \in \mathcal{S}, m \neq n} \frac{1}{N_S} \overline{y_{mi}}\right] \frac{\lambda L C_n \overline{\Sigma_i}}{1 - \lambda}\right) \tag{39c}$$

$$W_i = \frac{1}{\eta_i} \left\{\eta_{i0} W_{i0} + L \sum_n \sum_{m \in \mathcal{S}} \frac{1}{N_s} \overline{z_{mi}} C_m \overline{t_m^{-1}}\right\}$$
$$\eta_i = \eta_{i0} + L \sum_n \overline{z_{ni}}. \tag{39d}$$

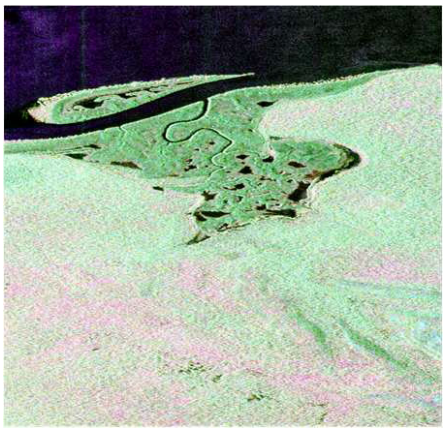
Compared with (31c)–(31d), the local correlation is learned by the dual constraints in (39c)–(39d). Finally, by locating the largest element of latent category-indicator variable z_n, the classification result label_n of nth pixel is calculated.

Algorithm 1 exhibits the complete inference process of the proposed DDPMM with dual constraints. First, all hyperparameters are initialized, and $r_{ni}^{(Z)}$ and $r_{ni}^{(Y)}$ are obtained by K-means algorithm. And then, the proposed DDPMM will be alternatively updated according to VBE-Step and VBM-Step to individually determine each parameter until reaching the maximum iteration number or the preset convergence value. Particularly, in the training stage, the local similarity is explored by the hyperparameters, and propagated to the latent variables Z and Y, respectively. The discriminative information included in polarimetric matrices and texture details is integrated in $\ln \rho$, where the components in two DPMMs can be aligned. Meanwhile, though DP could determine the component number automatically, the insignificant component which corresponding a very small value of mixing coefficient will also be annihilated. Finally, the proposed model outputs the classification result by (33).

III. Experimental Results

For quantitatively and qualitatively discussing the classification performance, four real PolSAR datasets are selected

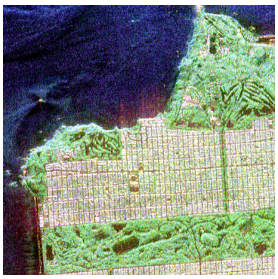
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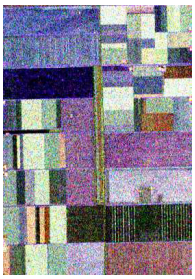
(a)



(b)



(c)



(d)

Fig. 2. Pauli RGB images of four test sites. (a) PolSAR image of the savanna area with the size of 792 × 827 was obtained by Nasa/JPL AIRSAR over the north of Kakadu NT, Australia. (b) This is a excerpt of the L-band PolSAR image with the size of 403 × 406 was acquired by Deutsches Zentrum für LuftBund Raumfahrt (DLR) ESAR over Oberpfaffenhofen, Germany. (c) Excerpt of the NASA/Jet Propulsion Laboratory (JPL) airborne synthetic aperture radar (AIRSAR) P-band Pauli RGB image over the San Francisco Bay, the image size of which is 451 × 456. (d) Excerpt of the L-band PolSAR image with the size of 394 × 276 was obtained by NASA/JPL AIRSAR over agriculture in Flevoland.

Algorithm 1 Proposed DDPMM Algorithm With Dual Constraints

Input: Multilook PolSAR data $\mathbf{C}$, parameters M, Sz, ϵ_m, ϵ_l, $\alpha_0^{(\phi)}$, $\alpha_0^{(v)}$, a_{ni0}, b_{ni0}, η_{i0}, W_{i0}, and $\boldsymbol{\lambda}$.
Initialize: $r_{ni}^{(Z)}$ and $r_{ni}^{(Y)}$ via K-means clustering.
Step 1: Train the proposed model:
Current iteration number $N_{ite} \leftarrow 0$ **repeat**
VBE-step: calculate $\ln \rho^{(Z)}$ and $\ln \rho^{(Y)}$ according to (37) and (38), respectively.
Component Alignment: generate $\ln \rho$ by $\ln \rho^{(Z)}$ and $\ln \rho^{(Y)}$, and then update $q^*(\mathbf{Z})$ and $q^*(\mathbf{Y})$.
VBM-step: update $q^*(\boldsymbol{v}), q^*(\boldsymbol{\phi}), q^*(\mathbf{t}|\mathbf{Y})$, and $q^*(\boldsymbol{\Sigma})$ according to (27)-(29), where their parameters are calculated by (39a)-(39d), respectively. After obtaining the updating $\mathbf{v}$ and $\boldsymbol{\phi}$, the corresponding mixing coefficients $\boldsymbol{\omega}(\mathbf{v})$ and $\boldsymbol{\lambda}(\boldsymbol{\phi})$ are acquired, such that the cluster number can be automatically determined.
Until: Convergence is attained with $c_v \leqslant \epsilon_l$ or $N_{ite} > I_{tr}$
Step 2: Determine the best M via annihilating the insignificant component with $\omega(v)_i \leqslant \epsilon_m$
Step 3: Determine the new $q^*(\mathbf{Z})$ of existing categories, and calculate the label of each pixel via (33).
Return: $label_n$

[see Fig. 2], which involve in diverse terrain types, ranging from the land agriculture, forest, seaside, and savanna to streets, rivers, and buildings. Meanwhile, different sensors are also be considered (i.e., DLR E-SAR, CONVAIR, and NASA/JPL AIRSAR) for verifying the effectiveness of the model we proposed. To facilitate and simplify the description in each subsequent experiment, we abbreviate these four datasets according to their place names as "*Savanna*," "*Oberpfaffenhofen*," "*SanFranciscoBay*," and "*Flevoland*," and they are all preprocessed by the refined Lee filter [53] on the "PolSARpro" software. Experimental environment is MATLAB R2022a language on Windows system with i7-10700K CPU, and 32-GB RAM.

A. Smoothness and Neighborhood Size

In this section, two groups of experiments are performed on the *Savanna* dataset for quantitatively investigating the effect of hyperparameters W_{i0} as well as neighborhood size Sz to the classification results. At the same time, following the [21], [45], and [52], the hyperparameters are empirically set as $\eta_{i0} = 5e0$, $\alpha_0^{(v)} = 1.1e\text{-}1$, $\alpha_0^{(\phi)} = 1.1e\text{-}1$, $a_{ni0} = 1.1e1$, for all n and i. The remaining parameters are set as $\boldsymbol{\lambda} = 5e\text{-}1$ and $M = 1e\text{-}1$, which are unified in all datasets. In particular, the initializing value of hyperparameter W_{i0} is the multiplication of an experiential value and an identity matrix $\mathbf{I}$, where the approximate range of the experiential value can be determined by averaging the MCC matrix's diagonal elements. This is because Σ follows the complex Wishart distribution, and the experiential value is dependent of the data due to the expectation of the complex Wisahrt distribution.

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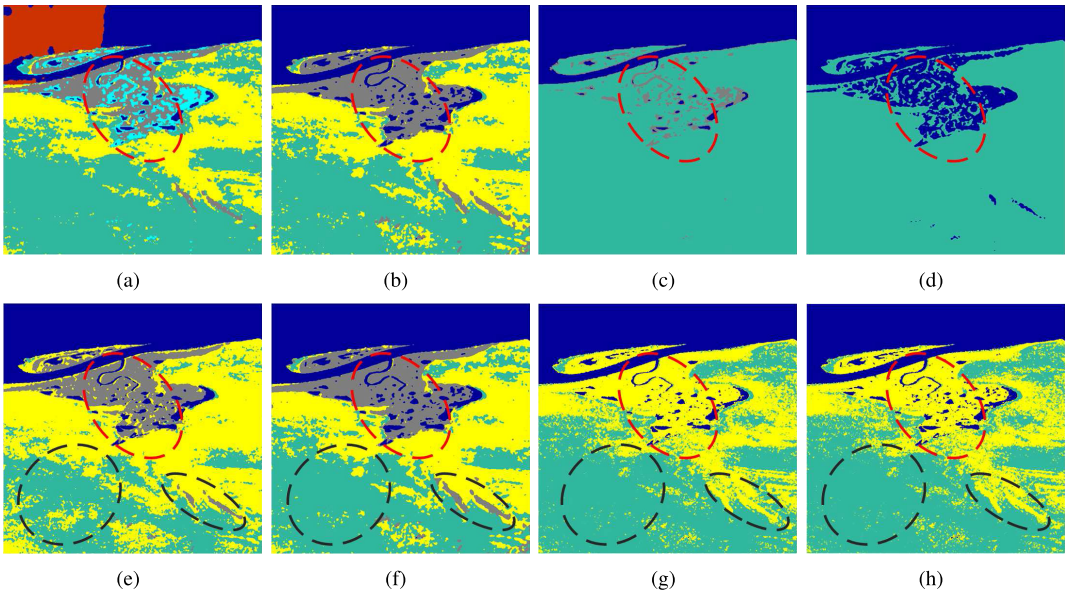


Fig. 3. Classification maps for the *Savanna* dataset. (a)–(d) Fix $Sz \times Sz = 5 \times 5$ and set the various value of W_{i0} with 1e-3 $\cdot$ **I**, 8e2 $\cdot$ **I**, 1e4 $\cdot$ **I**, and 1e5 $\cdot$ **I**. (e)–(h) Fix $W_{i0} = 8e2 \cdot \mathbf{I}$ and set the various value of $Sz \times Sz$ with 3×3, 5×5, 7×7, and 9×9, respectively.

Specifically, various initial values for hyperparameter W_{i0} are selected to be {1e-3 $\cdot$ **I**, 8e2 $\cdot$ **I**, 1e4 $\cdot$ **I**, 1e5 $\cdot$ **I**}, while the value of Sz is fixed as 5, and their corresponding resulting maps are presented in Fig. 3(a)–(d). Obviously, when a small value of W_{i0} is selected as 1e-3 $\cdot$ **I**, two new categories are recognized corresponding the light blue and the red parts in Fig. 3(a). In contrast, if the relatively large initial value of W_{i0} is determined with 1e5 $\cdot$ **I** or 1e4 $\cdot$ **I** as shown in Fig. 3(c) and (d), there are only two or three components existing, and the remaining components that have lower probabilities are eliminated. As for the pixels among red circle in Fig. 3(b), a better classification map with four components can be obtained by setting $W_{i0} = 8e2 \cdot \mathbf{I}$. Consequently, the four classification maps demonstrate that by adjusting the initial value of W_{i0}, the ultimate cluster number can be controlled to some extent, and selecting $W_{i0} = 8e2 \cdot \mathbf{I}$ is more applicable to the *Savanna* dataset. Significantly, the cluster numbers of these four maps in Fig. 3(a)–(d) are all less than the initialized value of M, which verifies the fact that the number of underlying categories can be determined via the proposed DDPMM automatically as long as M is large enough.

Subsequently, the classification results involved in various neighborhood scale $Sz \times Sz$ are given in Fig. 3(e)–(h), where the range of Sz is chosen from {3×3, 5×5, 7×7, 9×9} with fixing W_{i0} to be 8e2 $\cdot$ **I**. Their classification maps demonstrate that the hyperparameter Sz manipulates the incorporation of spatial context information. As presented by Fig. 3(e), when a small size with $Sz = 3$ is chosen for constructing a neighborhood pixel block, some gray mangroves pixels are identified as the yellow woodland pixels, especially in the red circle, and many green grass pixels in black circle are also misclassified as woodland. As such cases, though more detail information is reserved, the noise effect is not fully eliminated by the dual constraints in view of considering less neighborhood information. On the contrary, when Sz is set to be 7 or 9 of Fig. 3(g)–(h), much more local information is captured for smoothing labels, which, however, ignores the detail information to cause many misclassifications in the red circle. Therefore, a small Sz can make the resulting map persist the original structure by identifying more details, and setting a considerably larger Sz may lead to obtaining an over-smoothing classification result by focusing on the context information. Hence, to the *Savanna* dataset, its best value of Sz is 5. Furthermore, Table I lists the final expectation of each component in texture DPMM according to $\mathbb{E}[t_i^{-1}] = a_i/b_i$ in (32). Since the texture variable t is associated with the spatial fluctuation of the number of scatters (i.e., heterogeneity). It can be observed that all expectation is different from the initializations, which indicates the heterogeneity has been captured. By observing Table I, the grassland component has the largest value of expectation, indicating that the grassland area is more heterogeneous than other classes.

TABLE I
EXPECTATION OF EACH TEXTURE COMPONENT ON THE *Savanna* DATASET

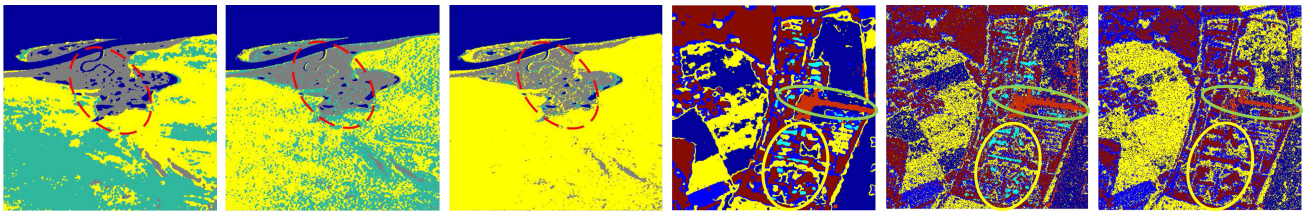
Class	Mangroves	Water	Woodland	Grassland
$\mathbb{E}[t_i^{-1}]$	0.9702	0.2823	0.5397	1.1559

B. Ablation Experiment

For validating the effectiveness of the proposed texture mixture strategy and spatial constraints, the proposed DDPMM without dual constraints (DDPMM-nonDC) and DDPMM without both dual constraints and texture mixture (DDPMM-nonDCTM) are utilized to conduct the ablation experiments, respectively, whose resulting maps are presented in Fig. 4.

Obviously, after omitting the local spatial constraints, the resulting map of DDPMM-nonDC cannot maintain good smoothness since DDPMM-nonDC becomes a pixel-level classification model rather than patch-level one. Besides,

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(a) (b) (c) (d) (e) (f)

Fig. 4. Ablation experiments of two sub-models. The first three classification maps correspond to the *Savanna* dataset. The last three classification maps correspond to the *Oberpfaffenhofen* dataset. (a) and (d) Classification results using the proposed DDPMM. (b) and (e) Classification results using the DDPMM-nonDC. (c) and (f) Classification results using the DDPMM-nonDCTM.

some detailed information cannot be fully learned either. For example, compared with the Pauli RGB image of the *Savanna* dataset in Fig. 2(a), many water pixels in the red circle are misidentified as mangroves or grassland pixels by DDPMM-nonDC in Fig. 4(b), and many incorrect woodland pixels are observed at lower left corner. Thus, imposing dual constraints in our model to incorporate local spatial context can effectively mitigate the effect of noise. When further removing the designed texture DPMM in Fig. 4(c), DPMM-nonDCTM misclassifies most grassland pixels in the whole map, since DPMM-nonDCTM degenerates to the general DPMM, and the texture details cannot be captured sufficiently. The above conclusion can also be confirmed by Fig. 4(d)–(f). When not considering the local similarity included in each pixel patch, DDPMM-nonDC cannot explicitly exhibit the building structure in the yellow circle, and DDPMM-nonDCTM is even unable to recognize the blue building pixels, because the single texture distribution is insufficient for the complex texture scene. Hence, by integrating the texture DPMM and dual spatial constraints, the proposed model can reach a significant classification improvement.

C. Analysis of Classification Performance

Four PolSAR datasets [see Fig. 2] with various complex scenarios are considered to validate the effectiveness of the proposed DDPMM, where Fig. 2(a) and (b) are unlabeled datasets, and Fig. 2(c) and (d) are labeled. For accurately evaluating the effectiveness of proposed DDPMM, we select six comparative algorithms, i.e., H/α-Wishart [16], CD-Wishart [54], $\mathcal{G}_p^0$MM [2], SVWMM [21], CK-HDRF [29], and SV$\mathcal{G}_p^0$MM [52], where the H/α-Wishart and CD-Wishart models are commonly-used baselines, $\mathcal{G}_p^0$MM and SV$\mathcal{G}_p^0$MM are the PM-based method with marginalized inverse Gamma texture variable, the SVWMM, CK-HDRF and SV$\mathcal{G}_p^0$MM are all the pairwise constraint-based classification methods. Considering that the CD-Wishart, the H/α-Wishart, and the CK-HDRF cannot automatically determine the number of clusters, their truncation level of M are set to be {4, 6, 4, 8} according to the number of terrain types in the *Savanna*, *Oberpfaffenhofen*, *SanFranciscoBay*, and *Flevoland* datasets, respectively. For the SVWMM, SV$\mathcal{G}_p^0$MM, and proposed DDPMM, their truncation level M is set to be 10 in view of the redundant components being merged with the nearest strategy when acquiring the classification labels. Since the parameter setting for the proposed DDPMM on the *Savanna* dataset has been given in Section III-A, the corresponding setting on another three datasets are listed in Table II.

TABLE II

PARAMETER SETTING ON THE *Oberpfaffenhofen*, *SanfranciscoBay*, AND *Flevoland* DATASETS

Parameters	*Oberpfaffenhofen*	*SanfranciscoBay*	*Flevoland*
ϵ	ϵ_m=1e-3, ϵ_l=1e-6		
Sz	5	5	5
W_{i0}	1.50e-2·$\mathbf{I}$	1.01e2·$\mathbf{I}$	2.45e-1·$\mathbf{I}$
η	5.00e0	5.00e0	5.00e0
$\alpha_0^{(v)}$	1.10e-1	1.15e-1	1.20e-1
$\alpha_0^{(\phi)}$	1.10e-1	1.15e-1	1.20e-1
a_{ni0}	1.10e1	1.04e1	1.03e1
b_{ni0}	1.00e1	9.40e0	9.30e0

1) Classification for Unlabeled Savanna and Oberpfaffenhofen Datasets:

a) Savanna dataset: According to Fig. 2(a), we can observe that *Savanna* dataset has four land covers: mangroves, water, woodland, and grassland (corresponding to the gray, blue, yellow, and green pixels in the first row of Fig. 5, respectively). The classification results by four unsupervised methods are exhibited in Fig. 5(a)–(d). In Fig. 5(a), CD-Wishart only recognizes mangroves, water, and grassland pixels, where the class "woodland" (i.e., yellow pixels) is not identified. The main reason is that the mixture strategy is not introduced for improving the statistical description. Although the SVWMM and SV$\mathcal{G}_p^0$MM in Fig. 5(b) and (c) can correctly identify the pixels in homogeneous areas, they cannot obtain the accurate classification maps in heterogeneous areas. For example, the grassland pixels on the left side of the island are misclassified as woodland class, and the water (blue) pixels marked by the red circle are also classified as mangroves pixels. This is due to the fact that the SVWMM doesn't introduce the PM assumption to construct the texture variable and SV$\mathcal{G}_p^0$MM doesn't fully consider the mixture texture strategy to capture complex texture details. For the resulting map in Fig. 5(d), the proposed DDPMM presents the accurate classification results especially of the water and mangroves pixels in the red circle. The result of our model verifies that DDPMM can capture texture information sufficiently and flexibly in view of designing the texture DPMM and incorporating the spatial context dynamically.

b) Oberpfaffenhofen dataset: According to Fig. 2(b), the *Oberpfaffenhofen* dataset includes six categories: woodland, bare field, crop 1, crop 2, roads, and buildings (corresponding

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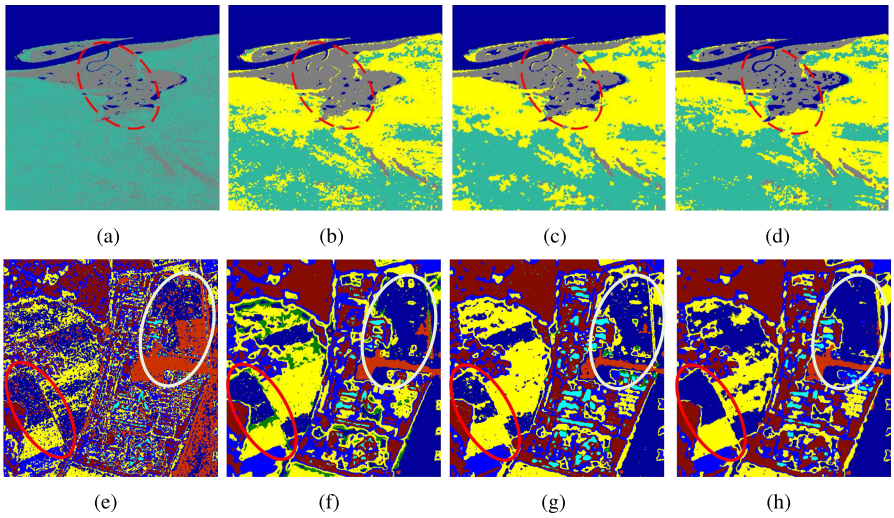


Fig. 5. Classification maps resulting of two datasets from the different methods. The first row corresponds to the *Savanna* dataset. The second row corresponds to the *Oberpfaffenhofen* dataset. (a) and (e) Classification results using the CD-Wishart method. (b) and (f) Classification results using the SVWMM. (c) and (g) Classification results using the SV$\mathcal{G}_p^0$MM. (d) and (h) Classification results using the proposed model.

TABLE III
AVERAGE ELAPSED TIME OF DIFFERENT UNSUPERVISED METHODS ON *Savanna* AND *Oberpfaffenhofen* DATASETS

Methods	Time(s)	
	Savanna	*Oberpfaffenhofen*
CD-Wishart	644.1	258.7
SVWMM	396.8	67.7
SV$\mathcal{G}_p^0$MM	388.6	60.3
DDPMM-nonDCTM	231.7	31.3
DDPMM-nonDC	266.8	43.2
Proposed Model	438.1	87.0

to the crimson, dark blue, gold, cyan, orange, and light blue pixels in the second row of Fig. 5, respectively). For Fig. 5(e) and (f), CD-Wishart generates miscellaneous labels in the homogeneous areas without considering the local spatial similarity, and SVWMM identifies an incorrect crop terrain due to the insufficient learning of texture information. SV$\mathcal{G}_p^0$MM cannot exhibit a clear structure of heterogeneous crop and building areas in Fig. 5(g), especially in the areas marked by red and white circles. By incorporating the local spatial similarity into the texture DPMM, the proposed DDPMM shows the more accurate and smoother classification result in Fig. 5(h). Thus, the proposed DDPMM has a significant classification improvement than other unsupervised methods.

Furthermore, Table III presents the average running time of compared methods (i.e., CD-Wishart, SVWMM, SV$\mathcal{G}_p^0$MM, DDPMM-nonDC, DDPMM-nonDCTM) and the proposed DDPMM on the *Savanna* and *Oberpfaffenhofen* datasets. From Table III, compared with the CD-Wishart, the proposed DDPMM takes a lower average elapsed time and shows higher efficiency. This is because the proposed method can automatically determine the component number, while a large truncation level (like 48) is initialized in the Chernoff-Wishart method, which spends much time to merge the redundant clusters according to the nearest strategy. Compared with SVWMM and SV$\mathcal{G}_p^0$MM, although the proposed DDPMM takes a little more time, the better classification results can be obtained by DDPMM due to the responsibility weight of local spatial constraints being dynamically updated for obtaining better classification results. In summary, though the proposed method needs more time, the classification results in Fig. 5 show the significant improvement compared to competing methods. Moreover, by removing the dual constraints and texture DPMM from the proposed model, the DDPMM-nonDC and DDPMM-nonDCTM's average elapsed times indicate that the local spatial constraints have higher overhead than the texture mixture strategy.

2) Classification for SanFranciscoBay Dataset: Fig. 6(a) exhibits the optical image of *SanFranciscoBay* dataset provided by Google Earth. According to Fig. 2(c), we can observe the *SanFranciscoBay* dataset has four land covers: field, ocean, vegetation, and urban (corresponding to the blue, yellow, green and orange pixels in Fig. 6, respectively), and the corresponding ground-truth map is shown in Fig. 6(b).

By observing Fig. 6(d)–(f), most of the ocean pixels in the image are accurately distinguished by the H/α-Wishart, CD-Wishart, and $\mathcal{G}_p^0$MM approaches, while massed of pixels of urban class in the white circle are recognized as vegetation class via both of them in view of the spatial context being neglected. Though the SVWMM and CK-HDRF in Fig. 6(e) and (h) introduce the local spatial information for obtaining the smoother resulting maps (e.g., the better results they calculated in the white circle), too much detail information is ignored. For example, the field pixels in two red circles are not identified, and the ocean areas in the purple circle of the upper right region of the image are also misjudged, this is because they do not introduce the texture variable. Compared with the SV$\mathcal{G}_p^0$MM in Fig. 6(i), since the texture variable in the proposed DDPMM is not marginalized, the texture information can be sufficiently captured by the unique texture vector, which allows the urban and field pixels at the white/red circle to include fewer misclassified points. Meanwhile, the clear boundaries are also acquired by the proposed DDPMM, especially for the pixels in the purple

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TABLE IV
ACCURACY ASSESSMENT BASED ON THE *SanfranciscoBay* DATASET

Class	Producer's Accuracy (%)						
	H/α-Wishart	CD-Wishart	$\mathcal{G}_p^0$MM	SVWMM	CK-HDRF	SV$\mathcal{G}_p^0$MM	Proposed Model
Field	91.63	82.50	66.79	83.02	74.30	95.47	**99.61**
Ocean	90.17	79.70	85.78	89.29	86.74	93.64	**95.78**
Urban	29.23	44.24	58.51	95.35	96.28	97.25	**98.52**
Vegetation	68.50	88.61	69.12	**95.90**	95.40	93.93	90.65
OA	66.73	69.87	73.34	92.32	91.15	94.90	**96.04**
κ	53.18	57.71	61.49	88.36	86.68	92.19	**93.85**

TABLE V
ACCURACY ASSESSMENT BASED ON THE *Flevoland* DATASET

Class	Producer's Accuracy (%)							
	H/α-Wishart	CD-Wishart	$\mathcal{G}_p^0$MM	DDPMM-nonDC	SVWMM	CK-HDRF	SV$\mathcal{G}_p^0$MM	Proposed Model
Rapeseed	63.58	89.78	75.59	88.77	97.06	99.01	99.04	**99.97**
Pea	91.79	94.41	89.44	95.25	99.47	**99.48**	97.74	97.84
Wheat	32.24	48.87	66.74	78.26	94.31	97.19	**97.99**	97.09
Lucerne	90.67	–	78.09	89.91	**95.50**	91.43	92.19	90.03
Barley	90.64	91.17	83.43	76.95	96.59	98.06	98.46	**98.64**
Beet	82.73	84.56	82.91	68.54	**98.96**	90.10	93.43	98.60
Potato	98.23	98.56	96.54	**98.97**	98.38	97.74	95.39	97.56
Bare Soil	94.87	95.40	92.46	94.34	96.40	98.28	97.55	**99.81**
OA	69.89	75.06	79.71	83.99	96.61	97.25	97.41	**97.83**
κ	65.17	70.42	75.76	80.55	95.84	96.61	96.80	**97.34**

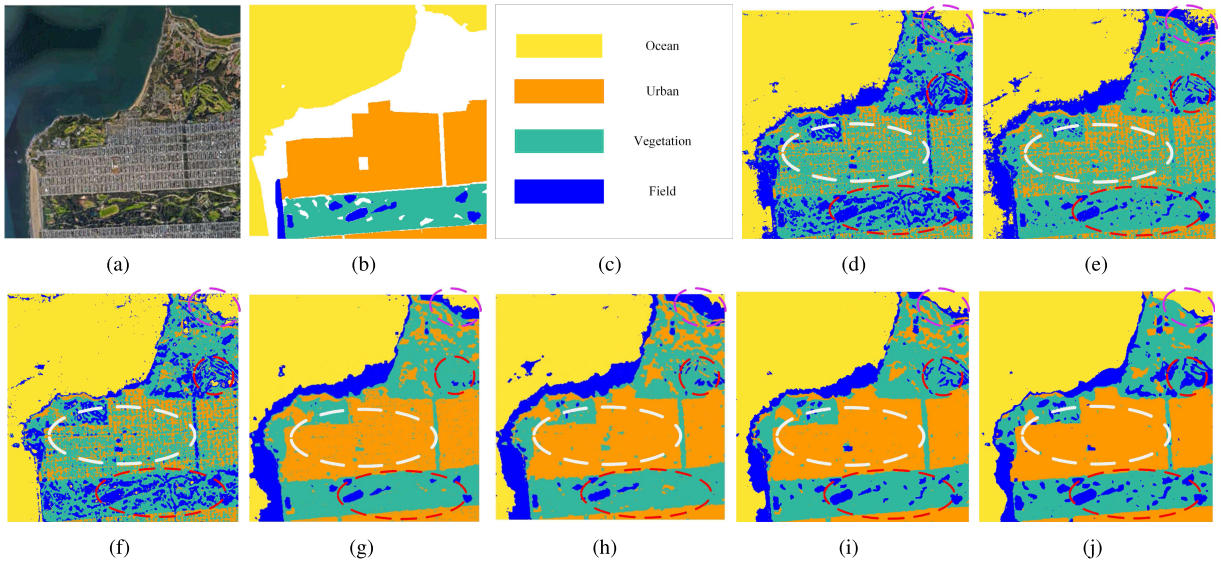


Fig. 6. *SanFranciscoBay* dataset and its classification maps resulting from the different methods. (a) Optical image from GoogleEarth. (b) Ground-truth map. (c) Color code. The resulting classification maps by employing (d) H/α-Wishart model, (e) CD-Wishart model, (f) $\mathcal{G}_p^0$MM model, (g) SVWMM, (h) CK-HDRF, (i) SV$\mathcal{G}_p^0$MM model, and (j) proposed model.

circles and at the junction of ocean and vegetation areas. Consequently, compared with all comparative methods, the proposed DDPMM obtains the fewest confused pixels and provides the clearest edge delineations by fully embedding the dual constraints into the PM framework. The highest values of κ coefficient and OA at the last column of Table IV also validate the validity of our DDPMM.

3) Classification for Flevoland Dataset: Fig. 7(a) and (b) present the ground truth map of the *Flevoland* dataset, and show the corresponding color map with eight types of crops. The classification maps generated by the proposed DDPMM as well as other comparative methods are shown in Fig. 7(c)–(j), and for the quantitative evaluation, their corresponding performance analyses are all summarized in Table V, including the OA, κ coefficient, and producer's accuracy. Due to the fact that the same crop types can be always found among a local region of the PolSAR image, the spatial context information will be significant to encourage the centered pixel to match

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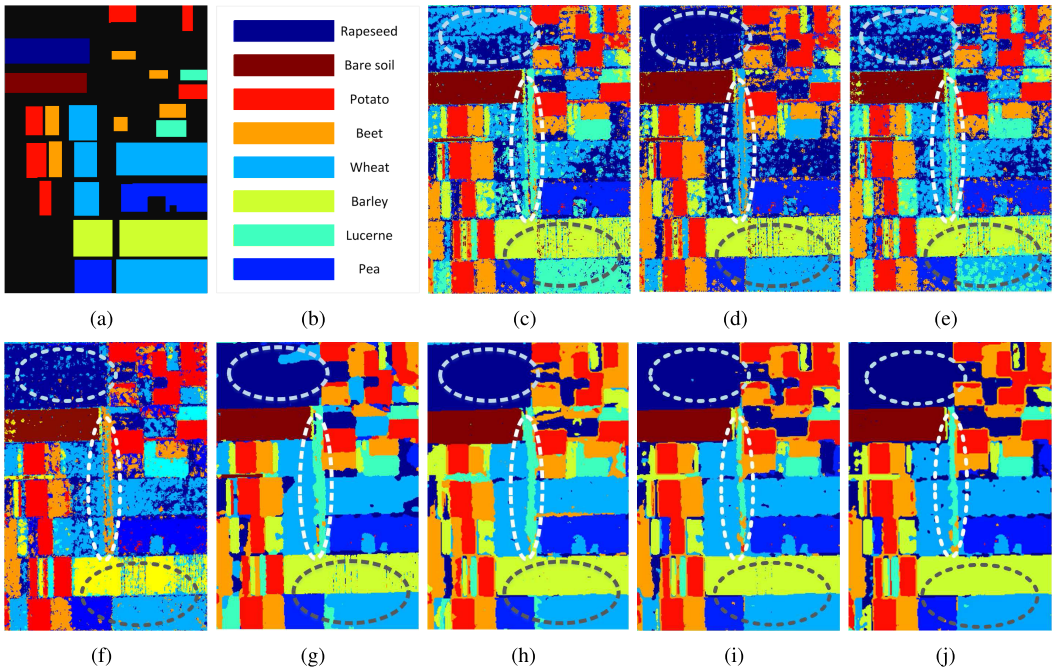


Fig. 7. *Flevoland* dataset and its classification maps resulting from the different methods. (a) Ground-truth map. (b) Color code. The resulting classification maps by employing (c) H/α-Wishart method, (d) CD-Wishart method, (e) $\mathcal{G}_p^0$MM, (f) DDPMM-nonDC, (g) SVWMM, (h) CK-HDRF model, (i) SV$\mathcal{G}_p^0$MM, and (j) proposed model.

the same labels with its adjacent pixels in a local pixel patch.

From Fig. 7(c)–(e), we can observe that without incorporating the spatial context, the H/α-Wishart, CD-Wishart, and $\mathcal{G}_p^0$MM cannot provide the accurate classification results with ground truth map, especially for the pixels marked by the white and brown circles. Particularly, compared with Fig. 7(e) and (f), though both $\mathcal{G}_p^0$MM and proposed DDPMM-nonDC employ the mixture model framework and introduce the texture variable, our DDPMM-nonDC acquires better classification results in the white and brown circles, which indicates the designed texture DPMM can more effectively distinguish those targets with similar scattering characteristics, like rapeseed and wheat areas. Hence, the proposed double mixture model has significant performance improvement than the widely-used single mixture model. The remaining four methods like SVWMM, CK-HDRF model, SV$\mathcal{G}_p^0$MM, and the proposed DDPMM, which leverage the local similarity measures, all persist the main terrain structures as ground truth map and produce less noise points in Fig. 7(g)–(j). However, according to the classification maps of SVWMM and CK-HDRF methods, many pixels at the junction of two terrain types, especially in the gray and the lower white circles, are identified as lucerne category, making the boundary be confused. Thus, it could be found that the SVWMM as well as the CK-HDRF is insufficient to process the extremely heterogeneous crop areas, due to the fact that texture detail information is exploited inadequately. Compared with SV$\mathcal{G}_p^0$MM, the proposed DDPMM can implement the accurate classification results, where fewer misclassification pixels in each rectangle crop areas are obtained (like rapeseed, barley, and Bare soil areas) and a clearer edge is presented by it. Therefore, the resulting map of the proposed DDPMM in Fig. 7(j) also verifies that integrating local spatial constraints into the model and updating them dynamically can generate better classification results. The conclusion can be confirmed via the highest OA and κ values in Table V.

IV. Conclusion

In this article, a novel unsupervised DDPMM under the variational Baysian framework is proposed for multilook PolSAR image classification. First, following the PM theory, the texture variable is introduced to deal with the heterogeneity in the complex scenes, and the statistical property of PolSAR observation data is modeled by the speckle-derived distribution. And then, the polarization DPMM and texture DPMM are specially designed to describe the polarimetric matrix and positive real-valued texture variable, respectively, thus making the generation procedure of the observation data be explicitly and sufficiently learned. Since each data point is associated with its own texture variable rather than sharing the same texture variable by marginalization operation, the additional flexibility is provided to learn the texture information. Simultaneously, instead of utilizing a single texture distribution, the texture DPMM represents the texture variable as a weighted sum of several densities, allowing the model to characterize more complex distribution profile of texture variable. Moreover, pairwise constraints based on the statistical representations of polarization and texture spaces are applied on the two DPMMs, respectively, for eliminating the noise influence. All parameters are determined and their closed-form updates are developed via the VI algorithm. Finally, experimental results on four real-world PolSAR datasets involved in various bands and sensors show that the proposed model

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has a better performance than some state-of-the-art statistical modeling-based classification methods.

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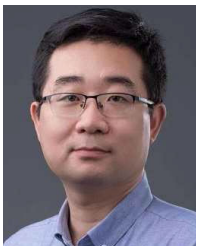
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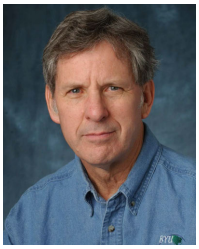
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